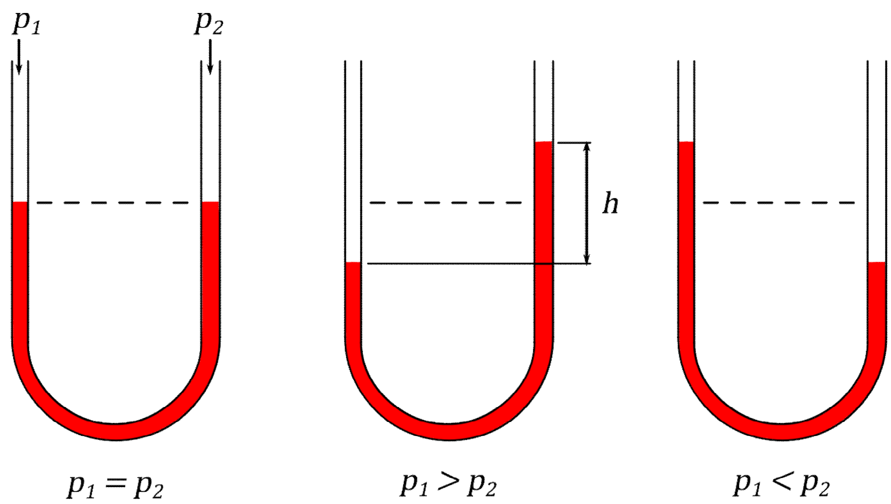
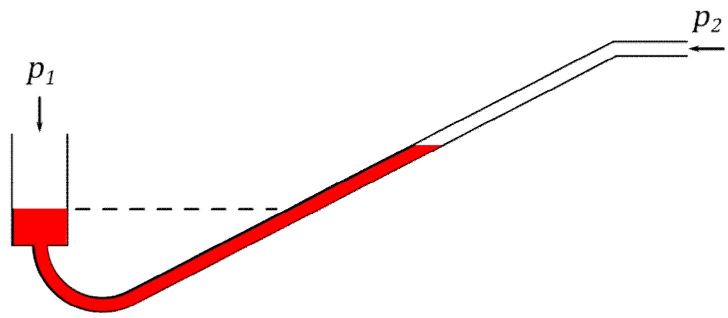
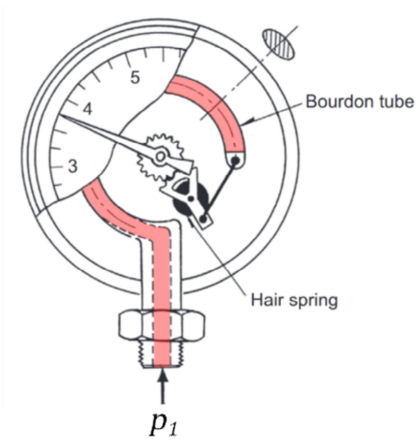


(a) U-tube manometer



(b) Inclined manometer

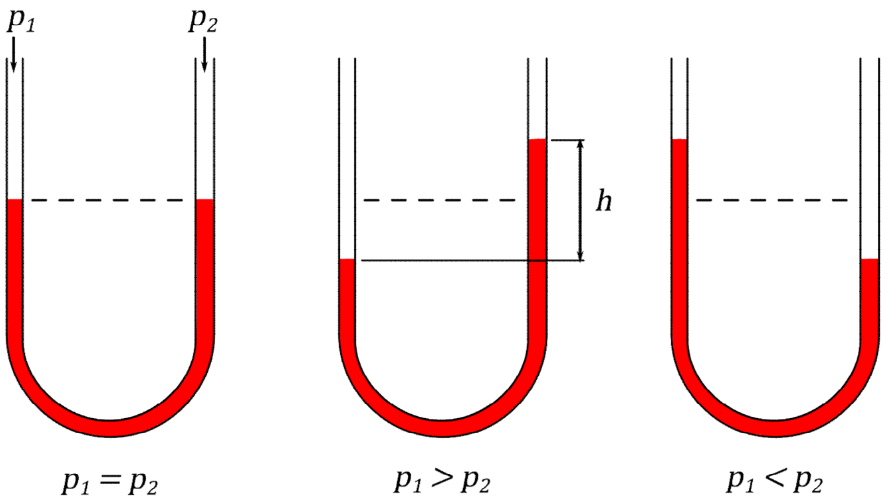


(c) Bourdon gauge, adapted from [8]

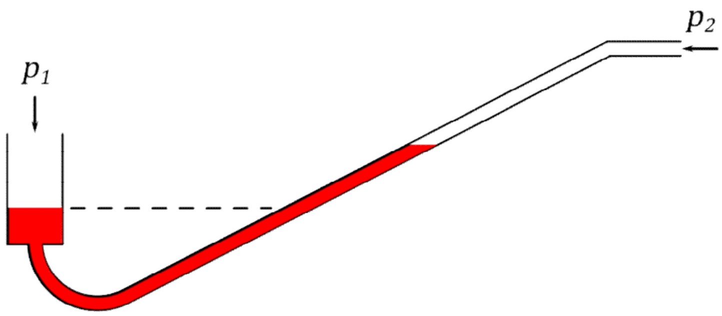
Figure 4 – Schematics of common Manometers

<u>Bourdon Gauge</u>

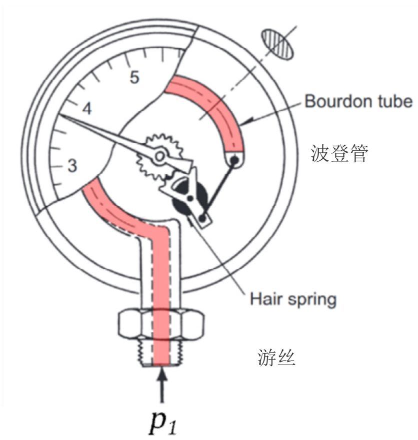
Shown in Figure 4(c) perhaps the most common manometer is the Bourdon tube gauge, named after their inventor. Readily available as both analogue or digital devices, with either absolute or gauge pressure scales, these devices feature a coiled, elliptical cross-section tube that tends to elongate and straighten when pressurised. This allows



(a) U 型管压力计 (U-tube manometer)



(b) 倾斜式压力计 (Inclined manometer)



(c) 波登管压力表 (Bourdon gauge)，改编自 [8] (adapted from [8])

Figure 4 – 常见压力计示意图（Schematics of common Manometers）

波登管压力表 (Bourdon Gauge)

如图 4(c) 所示，最常见的压力计 (manometer) 可能是以发明者命名的波登管压力表 (Bourdon tube gauge)。
这类装置有模拟式 (analogue) 或数字式 (digital)，压力刻度 (pressure scales) 可为绝对压力 (absolute pressure) 或表压 (gauge pressure)。
它们采用盘绕的椭圆截面管 (coiled, elliptical cross-section tube)，受压 (pressurised) 时会趋向伸长并变直。
这使得

Table 1 – Comparison of Temperature Transducers

Type	Cost	Output	Linearity	Temp. Range (ºC)	Individual Sensor Range (K)	Readily Achievable Accuracy (K)
TC	medium	low	reasonable	-250 to 2500	≈1000	±1.0
PRT	high	low	good	-200 to 850	≈500	±0.1
Thermistor	low	high	poor	-50 to 250	≈100	±1.0

3.3 Introduction to Rotational Speed Measurement Devices

The measurement of rotational speed, while not as commonplace as pressure or temperature measurements, is still widespread. It has application to rotating machinery such as internal combustion engines, turbomachinery (including wind turbines and jet engines) and aircraft attitude and motion.

For machines that operate only in 1-degree-of-rotational-freedom the speed is usually stated in units of **revolutions per minute (rpm)** rather than in Hz. In comparison, for aircraft the rotational motion occurs in 3-degrees-of-freedom (3-DoF) and is typically measured in radians per second (rad/s) by means of a **gyroscope**. In the interest of brevity, we will constrain this discussion to just measurements of 1-DoF rotation. A discussion of multi-DoF measurements using gyroscopes is left for future courses and study.

Today, most speed measuring systems use **pulse counting methods.** In these methods either an inductive or optical probe senses the passage of a feature as it rotates past and produces a corresponding train of pulses in time. The feature being detected depends on the specific design and application but could be the teeth on a gear wheel (or even the flats of a hexagonal nut), or an etched or anodized pattern on a shaft (for optical detectors). Pulse counting can either be digital (accurate) or analogue (less accurate) as follows –

1. Digital Pulse Counting: in these methods the time between pulses is recorded using a **quartz crystal oscillator** which typically have an accurate to better than ±0.02 %. The pulse time is then scaled to rpm digitally.
2. Analogue Pulse Counting: in these methods a dedicated integrated circuit is used which averages the pulses and converts the pulse frequency to a voltage. An example such device is the *Texas Instruments LM2917-N*. The voltage output

Table 1 - 温度传感器对比 (Comparison of Temperature Transducers)

类型 **(Type)**	成本 **(Cost)**	输出 **(Output)**	线性度 **(Linearity)**	温度范围 **(Temp. Range, °C)**	单个传感器量程 **(Individual Sensor)** **(Range, K)**	易实现精度 **(Readily Achievable)** **(Accuracy, K)**
热电偶 (TC)	中等 (medium)	低 (low)	合理 (reasonable)	-250 to 2500	≈ 1000	± 1.0
铂电阻 (PRT)	高 (high)	低 (low)	好 (good)	-200 to 850	≈ 500	± 0.1
热敏电阻 (Thermistor)	低 (low)	高 (high)	差 (poor)	-50 to 250	≈ 100	± 1.0

3.3 旋转速度测量装置简介 (Introduction to Rotational Speed Measurement Devices)

旋转速度 (rotational speed) 的测量虽然不如压力 (pressure) 或温度 (temperature) 测量那样常见，但仍然应用广泛。它适用于内燃机 (internal combustion engines)、涡轮机械 (turbomachinery，包括风力涡轮机和喷气发动机) 以及飞机姿态与运动 (aircraft attitude and motion) 等旋转机械。

对于只有一个旋转自由度 (1-degree-of-rotational-freedom) 的机器，速度通常用每分钟转数 (revolutions per minute, rpm) 表示，而不是用 Hz。相比之下，飞机的旋转运动具有三个自由度 (3-degrees-of-freedom, 3-DoF)，通常通过陀螺仪 (gyroscope) 以弧度每秒 (rad/s) 测量。为简洁起见，这里只讨论一自由度 (1-DoF) 旋转测量；多自由度陀螺仪测量留待后续课程学习。

如今，大多数速度测量系统使用脉冲计数方法 (pulse counting methods)。在这类方法中，感应式或光学探头 (inductive or optical probe) 检测旋转特征经过探头的时刻，并产生相应的时间脉冲列 (train of pulses)。被检测特征取决于具体设计和应用，可以是齿轮齿、六角螺母平面，或轴上的蚀刻/阳极氧化图案。

between the piston and the gas is found from $\oint pdv$, where $\oint$ denotes the integration over an engine cycle (720º crank angle for these 4 stroke engines).

4.3.2 Mechanical Efficiency

The **mechanical efficiency** of the engine is, like all efficiencies, a ratio of energy input vs energy output. As the engine operates at a certain (rotational) speed, it is more convenient to use **power** (energy per time) rather than energy for the definition. Specifically, the mechanical efficiency is defined as the ratio of **brake power** ($\dot{W}_b$) delivered by the engine to the **indicated power** ($\dot{W}_i$)–

$$\eta_{mech} = \frac{\dot{W}_b}{\dot{W}_i} \qquad (4.7)$$

These terms are defined as follows. Firstly, the work done by the gas on the piston over a single cycle is known as the **indicated work** and is given by –

$$W_i = \oint p\, dV$$

This is work per cycle but can be converted into **indicated power**, $\dot{W}_i$, by multiplying by the number of cycles per second (N'):

$$\dot{W}_i = W_i\, N' \qquad (4.8)$$

where for a four-stroke engine with n cylinders running at N rpm, in which a cycle is completed in 2 revolutions, we can define N' as:

$$N' = \frac{N\, n}{120} \qquad (4.9)$$

The **brake power** is the product of the engine torque at the flywheel (or dynamometer) and the angular velocity in rad/s –

$$\dot{W}_b = T\,\omega \qquad (4.10)$$

The difference between indicated work and brake work is the **power dissipated as friction**, that is –

$$\dot{W}_f = \dot{W}_i - \dot{W}_b \qquad (4.11)$$

活塞 (piston) 与气体 (gas) 之间的功由 $\oint p\,dV$ 给出，其中 $\oint$ 表示对一个发动机循环 (engine cycle) 积分；对这些四冲程发动机 (4 stroke engines) 而言，一个循环对应 $720°$ 曲轴转角 (crank angle)。

4.3.2 机械效率 (Mechanical Efficiency)

机械效率 (mechanical efficiency) 与所有效率一样，是能量输入 (energy input) 与能量输出 (energy output) 的比值。由于发动机以一定的转速 (rotational speed) 运行，用功率 (power)，也就是单位时间能量 (energy per time)，来定义会比直接用能量更方便。具体而言，机械效率定义为发动机输出的制动功率 (brake power) $(\dot{W}_b)$ 与指示功率 (indicated power) $(\dot{W}_i)$ 之比：

$$\eta_{\text{mech}} = \frac{\dot{W}_b}{\dot{W}_i} \tag{4.7}$$

这些量定义如下。首先，气体在单个循环内对活塞所做的功称为指示功 (indicated work)，可写为：

$$W_i = \oint p\,dV$$

这是每循环的功；将其乘以每秒循环数 (N')，即可得到指示功率 (indicated power) $\dot{W}_i$：

$$\dot{W}_i = W_i N' \tag{4.8}$$

对于一台以 N rpm 运转、具有 n 个气缸 (cylinders) 的四冲程发动机 (four-stroke engine)，每个循环需要 2 转，因此可定义 N' 为：

$$N' = \frac{N\,n}{120} \tag{4.9}$$

制动功率 (brake power) 等于飞轮 (flywheel) 或测功机 (dynamometer) 处的发动机扭矩 (torque) 与角速度 (angular velocity, rad/s) 的乘积：

$$\dot{W}_b = T\omega \tag{4.10}$$

指示功与制动功之间的差值，就是作为摩擦 (friction) 耗散掉的功率：

$$\dot{W}_f = \dot{W}_i - \dot{W}_b \tag{4.11}$$